

CONTROLS SYSTEMS

Project 01

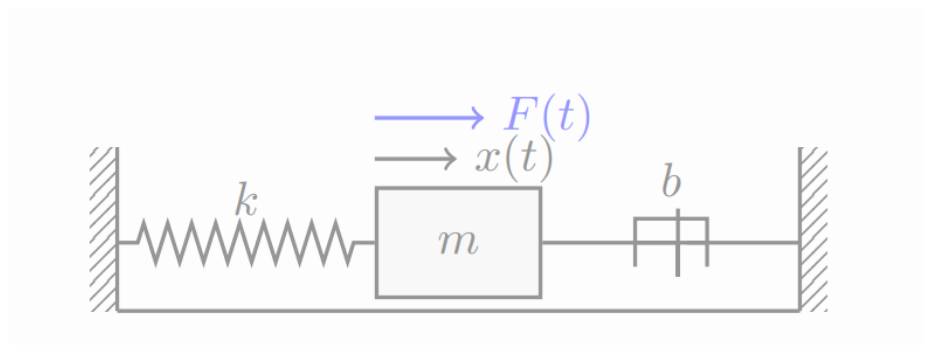
Dynamic Analysis of Mass–Spring–Damper Systems Using MATLAB

Designed, Developed and Documented by

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MATLAB R2024b • Control System Toolbox • Aerospace Engineering

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Abstract

This project presents the modeling and dynamic analysis of six mass–spring–damper system configurations using MATLAB. Transfer functions were derived for each configuration from the governing equations of motion, and the system behavior was investigated through step and impulse response analysis. Pole locations were examined to study system stability and dynamic characteristics, while transient performance specifications were evaluated for the complete mass–spring–damper system. Furthermore, the numerical solution obtained using the MATLAB ODE45 solver was compared with the transfer-function-based response to validate the mathematical model. The results demonstrate the influence of mass, spring stiffness, and damping coefficient on system dynamics and provide a fundamental understanding of second-order control systems relevant to aerospace engineering and future controller design.

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1 Objectives

The primary objective of this project was to develop a fundamental understanding of second-order dynamic systems through the mathematical modeling and simulation of various mass-spring-damper configurations using MATLAB. The project aimed to bridge the gap between theoretical control system concepts and their practical implementation by deriving mathematical models from Newton's Second Law, converting them into transfer functions using the Laplace Transform, and analyzing their dynamic behavior under different operating conditions.

Six different mechanical system configurations were investigated to study how changes in physical parameters influence system performance. Step and impulse responses were obtained for each configuration to observe the transient and steady-state characteristics of the systems. Furthermore, the locations of system poles, performance specifications, and numerical solutions obtained using the ODE45 solver were analyzed to verify the accuracy of the mathematical models.

The project also aimed to develop intuition regarding the influence of mass, damping, and stiffness on system stability, oscillations, settling time, rise time, overshoot, and natural frequency. This understanding serves as the foundation for future work involving controller design, state-space analysis, frequency-domain techniques, and aerospace flight control systems.

2 Introduction

Control systems are an essential part of modern engineering and are used wherever the behavior of a physical system must be monitored, regulated, or controlled to achieve a desired output. They are found in numerous engineering applications including aircraft flight control, spacecraft attitude stabilization, automotive cruise control, robotic manipulators, industrial automation, and vibration suppression systems.

Many practical engineering systems exhibit dynamic behavior, meaning that their outputs do not respond instantaneously to changes in the input. Instead, they require a finite amount of time to react because they store and dissipate energy. The analysis of such systems forms one of the most important areas of control engineering.

Among all dynamic systems, the mass-spring-damper system is considered one of the most fundamental mathematical models. Although physically simple, it accurately represents the dynamic behavior of many real engineering systems. Aircraft landing gear, vehicle suspension systems, vibration isolation mechanisms, satellite appendages, robotic joints, and structural vibration absorbers can all be approximated using mass-spring-damper models under suitable assumptions.

The dynamic response of a mass-spring-damper system depends primarily on three physical parameters: the mass, which provides inertia; the spring, which stores potential energy and generates restoring forces; and the damper, which dissipates mechanical energy and reduces oscillations. By varying these parameters, engineers can modify system performance to satisfy specific design requirements.

This project investigates six different mass-spring-damper configurations in order to understand how different physical arrangements influence system behavior. Mathematical models were derived using Newton's Second Law and transformed into transfer functions through the Laplace Transform. The resulting systems were analyzed using MATLAB to study their step responses, impulse responses, pole locations, and transient characteristics. Numerical solutions obtained through the ODE45 solver were also compared with analytical transfer-function responses to verify the correctness of the models.

The knowledge developed throughout this project provides a strong foundation for more advanced topics in control engineering, including feedback control, PID controller design, state-space modeling, optimal control, and aerospace flight control systems.

3 System Description

The mechanical systems investigated in this project consist of rigid masses connected to linear springs and viscous dampers in various configurations. An external force is applied to the mass, causing it to move along a single axis. The resulting displacement of the mass is considered the output of the system.

The spring generates a restoring force that always acts opposite to the direction of displacement according to Hooke's Law. This restoring force tends to return the mass toward its equilibrium position. The damper produces a resistive force proportional to the velocity of the mass, thereby dissipating kinetic energy and reducing oscillatory motion. The mass contributes inertia, resisting changes in motion according to Newton's Second Law.

To investigate the influence of different physical arrangements, six different system configurations were developed. These include systems containing only springs, only dampers, combinations of multiple springs or dampers, and the complete mass-spring-damper system. Each configuration represents a different dynamic behavior and allows the influence of individual physical parameters to be isolated and studied.

The responses of these systems were analyzed using both step and impulse inputs. The step input represents a force that is applied and maintained, whereas the impulse input represents a force applied over a very short duration and then removed. These two standard test signals provide valuable insight into the transient and steady-state behavior of dynamic systems.

4 Assumptions

To simplify the mathematical modeling and enable analytical solutions, the following assumptions were made throughout the project:

- The motion of the mass is restricted to one dimension.
- The spring obeys Hooke's Law and behaves as a perfectly linear spring.
- The damper is modeled as a linear viscous damper whose force is directly proportional to velocity.
- The mass is considered rigid and remains constant throughout each simulation.
- The effects of gravity are neglected because they do not influence the relative dynamic response under the chosen coordinate system.
- Friction other than the modeled viscous damping is ignored.
- The system parameters remain constant during each individual simulation.
- External disturbances and noise measurement are neglected.
- Initial displacement and velocity are assumed to be zero unless otherwise specified.
- The system is assumed to operate within its linear region, allowing the use of Laplace Transform techniques and transfer-function analysis.

These assumptions allow the systems to be modeled as linear time-invariant (LTI) systems, making them suitable for classical control analysis using MATLAB

5 Mathematical Modeling

5.1 Mathematical Modeling

The first step in analyzing any dynamic system is the development of an accurate mathematical model that represents its physical behavior. A mathematical model establishes the relationship between the external input applied to the system and the resulting output. In control engineering, this model serves as the foundation for stability analysis, controller design, performance evaluation, and simulation.

The mass-spring-damper system is governed by **Newton's Second Law of Motion**, which states that the sum of all external forces acting on a body is equal to the product of its mass and acceleration.

$$\sum F = m\ddot{x}$$

where

- m = Mass (kg)
- $x(t)$ = Displacement (m)
- $\dot{x}(t)$ = Velocity (m/s)
- $\ddot{x}(t)$ = Acceleration (m/s²)

In the complete mass-spring-damper system, three primary forces act on the mass:

Applied Force

The applied force, denoted by $F(t)$, is the external input that causes the mass to move. In this project, both step and impulse forces were used to investigate the dynamic response of the system.

Spring Force

According to Hooke's Law, the spring generates a restoring force proportional to the displacement of the mass from its equilibrium position.

$$F_s = kx$$

where

- k is the spring stiffness (N/m)

The spring always acts opposite to the direction of displacement and attempts to restore the mass to its equilibrium position.

Damping Force

The viscous damper produces a force proportional to the velocity of the mass.

$$F_d = b\dot{x}$$

where

- b is the damping coefficient (N·s/m)

Unlike the spring, the damper does not store energy. Instead, it dissipates mechanical energy, reducing oscillations and improving system stability.

5.2 Governing Differential Equation

Applying Newton's Second Law to the complete mass-spring-damper system gives

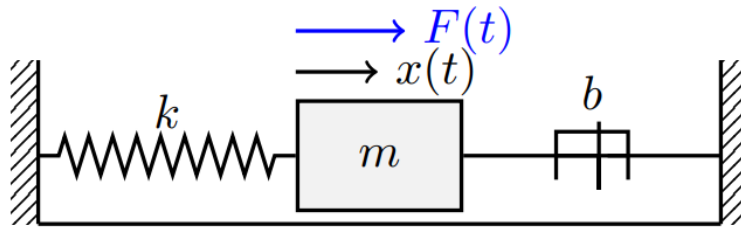


Figure 1 Governing equation diagram

Free Body Diagram

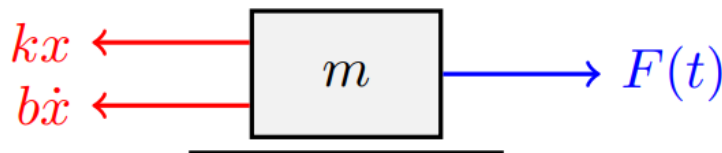


Figure 2 Free Body Diagram — full system

$$F(t) - b\dot{x} - kx = m\ddot{x}$$

Rearranging,

$$m\ddot{x} + b\dot{x} + kx = F(t)$$

This second-order differential equation completely describes the dynamics of the system.

- Each term has a clear physical interpretation:
- $m\ddot{x}$ represents the inertial force produced by the mass.
- $b\dot{x}$ represents the energy dissipated by the damper.
- kx represents the restoring force generated by the spring.
- $F(t)$ represents the external force acting on the system.

The relative magnitudes of these three physical parameters determine whether the system responds quickly, slowly, or with oscillations.

5.3 Transfer Function Development

Although differential equations accurately describe system dynamics, transfer functions provide a much more convenient representation for control system analysis. Transfer functions allow engineers to investigate system stability, transient response, steady-state response, and controller design using algebraic methods rather than solving differential equations directly.

Taking the Laplace Transform of the governing equation while assuming zero initial conditions,

$$ms^2X(s) + bsX(s) + kX(s) = F(s)$$

Factoring out $X(s)$,

$$X(s)(ms^2 + bs + k) = F(s)$$

Therefore, the transfer function relating displacement to the applied force becomes

$$G(s) = \frac{X(s)}{F(s)} = \frac{1}{ms^2 + bs + k}$$

This transfer function represents the complete mass-spring-damper system and serves as the fundamental model for most second-order mechanical control systems.

5.4 Development of Six System Configurations

To investigate the influence of individual physical elements on system dynamics, six different mechanical configurations were developed and analyzed throughout this project.

5.4.1 Case 01 – Mass-Spring System

In this configuration, damping was neglected, leaving only the mass and spring.

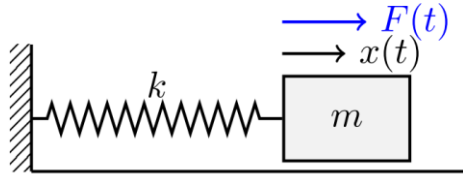


Figure 3 Case 01 system diagram

Free Body Diagram

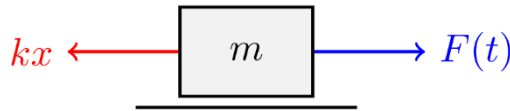


Figure 4 Free Body Diagram — Case 01

$$G(s) = \frac{1}{ms^2 + k}$$

This system stores mechanical energy but does not dissipate it. Consequently, oscillatory behavior dominates the response.

5.4.2 Case 02 – Mass-Damper System

In this case, the spring was removed and only the damper resisted motion.

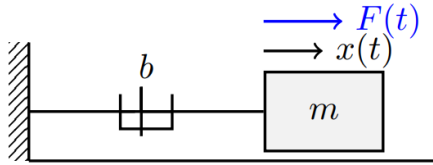


Figure 5 Case 02 system diagram

Free Body Diagram

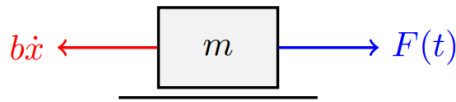


Figure 6 Free Body Diagram — Case 02

$$G(s) = \frac{1}{ms^2 + bs}$$

Since no restoring force exists, the system does not oscillate. Instead, the damping force gradually reduces the velocity of the mass until it reaches equilibrium.

5.4.3 Case 03 – Mass with Two Springs

Two identical springs were attached to the mass.

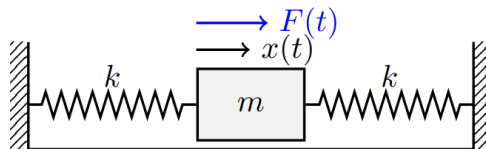


Figure 7 Case 03 system diagram

Free Body Diagram

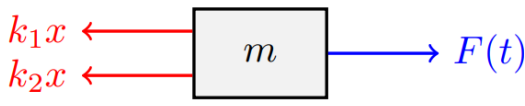


Figure 8 Free Body Diagram — Case 03

The equivalent stiffness becomes

$$k_{eq} = 2k$$

leading to

$$G(s) = \frac{1}{ms^2 + 2k}$$

The increased stiffness causes the system to oscillate at a higher natural frequency compared to the single-spring configuration.

5.4.4 Case 04 – Mass with Two Dampers

Two identical dampers were connected to the mass.

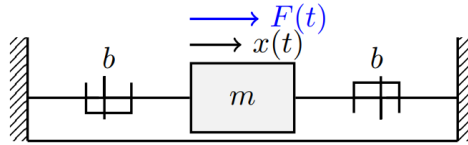


Figure 9 Case 04 system diagram

Free Body Diagram

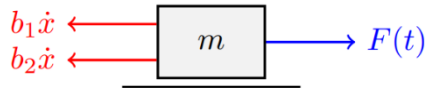


Figure 10 Free Body Diagram — Case 04

The equivalent damping coefficient becomes

$$b_{eq} = 2b$$

yielding

$$G(s) = \frac{1}{ms^2 + 2bs}$$

The additional damping increases energy dissipation and allows the system to settle more rapidly.

5.4.5 Case 05 – Spring and Damper on the Same Side

A spring and a damper were connected in parallel between the wall and the mass.

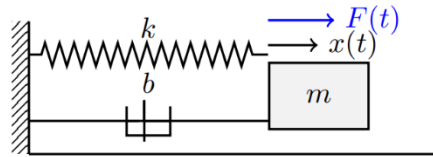


Figure 11 Case 05 system diagram

Free Body Diagram

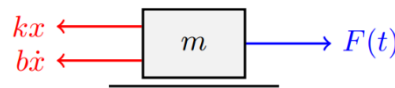


Figure 12 Free Body Diagram — Case 05

The governing transfer function becomes

$$G(s) = \frac{1}{ms^2 + bs + k}$$

This configuration represents the classical second-order control system encountered throughout mechanical and aerospace engineering.

5.4.6 Case 06 – Spring and Damper on Opposite Sides

The spring and damper were positioned on opposite sides of the mass.

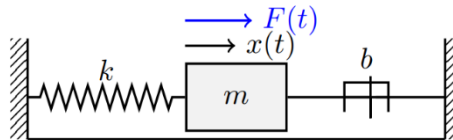


Figure 13 Case 06 system diagram

Free Body Diagram

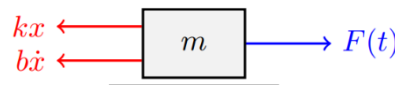


Figure 14 Free Body Diagram — Case 06

Although the physical arrangement differs from Case 05, both elements continue to oppose the motion of the mass. Consequently, the governing differential equation remains identical, producing the same transfer function,

$$G(s) = \frac{1}{ms^2 + bs + k}$$

As a result, Cases 05 and 06 exhibit identical mathematical behavior despite their different physical layouts.

5.5 Purpose of Parameter Variation

After deriving the mathematical models, the values of mass, spring stiffness, and damping coefficient were varied systematically to investigate their influence on system performance.

Instead of changing all parameters simultaneously, only one parameter was modified at a time whenever possible. This approach made it possible to isolate the contribution of each physical parameter and understand its individual effect on the transient and steady-state characteristics of the system.

The responses obtained from these variations form the basis of the analysis presented in the following sections, where step responses, impulse responses, pole locations, and performance specifications are discussed in detail.

6 MATLAB Implementation

6.1 MATLAB Implementation

Following the development of the mathematical models and transfer functions, MATLAB R2024b together with the Control System Toolbox was used to simulate and analyze the dynamic behavior of each system configuration. MATLAB provides a powerful computational environment for modeling linear dynamic systems and allows engineers to study both transient and steady-state responses without solving the governing differential equations manually.

Each of the six system configurations was implemented by defining its corresponding transfer function using the **Transfer Function (tf)** command. Once mathematical models were established, standard control system analysis tools were employed to investigate the dynamic characteristics of each case under different operating conditions.

The analysis primarily focused on observing the response of each system to standard test inputs, determining system stability, locating system poles, evaluating transient performance, and verifying analytical results through numerical solutions.

6.1.1 MATLAB Functions Used

Several built-in MATLAB functions from the Control System Toolbox were used throughout this project. Each function served a specific purpose in the overall analysis.

6.1.1.1 Transfer Function (tf)

The tf function was used to construct mathematical models directly from the derived transfer functions. The numerator and denominator coefficients corresponding to each system configuration were entered into MATLAB to create Linear Time-Invariant (LTI) models suitable for analysis.

```
num = 1;  
den = [m b k];  
G = tf(num,den);
```

This command generates the transfer function

$$G(s) = \frac{1}{ms^2 + bs + k}$$

which forms the basis for all subsequent simulations.

6.1.1.2 Step Response (step)

The step function was used to study the response of each system when subjected to a unit step input. This represents the application of a constant force that remains active throughout the simulation.

The response provides valuable information regarding

- Rise Time
- Peak Time
- Overshoot
- Settling Time
- Steady State Value
- Overall Stability

By comparing the step responses of different configurations, the influence of mass, damping, and spring stiffness on system performance was investigated.

6.1.1.3 Impulse Response (impulse)

The impulse function was used to analyze the response of each system to a unit impulse input. Unlike the step response, the impulse input represents a force applied over an extremely short duration before being removed.

Impulse responses reveal the natural dynamics of the system and are particularly useful for observing oscillatory behavior, damping characteristics, and energy dissipation.

6.1.1.4 Pole Calculation (pole)

The pole command was used to determine the locations of the poles of each transfer function.

```
p = pole(G);
```

The pole locations provide direct insight into system stability and transient behavior. Systems with complex conjugate poles exhibit oscillatory responses, whereas real poles typically produce non-oscillatory behavior. Throughout the project, the pole locations were compared with the corresponding response plots to establish the relationship between mathematical models and physical system behavior.

6.1.1.5 Performance Analysis (stepinfo)

The `stepinfo` function was employed to calculate important transient response specifications automatically.

The calculated parameters include

- Rise Time
- Peak Time
- Maximum Overshoot
- Settling Time
- Peak Value
- Steady-State Value

These numerical values were used to support the graphical observations obtained from the step response plots and to compare the performance of different system configurations quantitatively.

Note

The stepinfo function was applied only to the complete mass-spring-damper system (Case 5). The earlier configurations, such as the mass-spring and mass-damper systems, do not satisfy the assumptions required for meaningful transient performance analysis. For example, the undamped mass-spring system does not settle to a finite steady-state value, while the mass-damper system lacks the oscillatory characteristics of a standard second-order system. Therefore, performance specifications such as settling time, overshoot, and rise time were evaluated only for the complete second-order system.

6.1.1.6 Numerical Solution (ode45)

To verify the analytical transfer-function approach, MATLAB's ode45 solver was applied to the complete mass-spring-damper system.

The governing second-order differential equation was rewritten as a system of first-order differential equations before being solved numerically.

Unlike transfer-function analysis, which assumes a linear system represented in the Laplace domain, ode45 directly integrates the governing differential equations in the time domain. Comparing the numerical and analytical solutions provided additional confidence in the correctness of the developed mathematical model.

6.2 Simulation Procedure

The implementation of each case followed a common sequence to ensure consistency throughout the project.

1. The physical system was represented mathematically using Newton's Second Law.
2. The governing differential equation was transformed into the Laplace domain.
3. The transfer function corresponding to each configuration was developed.
4. The transfer function was implemented in MATLAB using the **tf** command.
5. Step and impulse responses were generated for each system.
6. Pole locations were calculated to investigate stability.
7. Performance specifications were evaluated using stepinfo where applicable.

8. The complete mass-spring-damper system was further verified using the ode45 numerical solver.
9. The responses obtained from different parameter variations were compared to identify the influence of mass, damping, and spring stiffness on system performance.

6.3 Parameter Variation Strategy

To understand the influence of each physical parameter independently, only one parameter was varied at a time while the remaining parameters were held constant whenever possible.

The investigation included variations in

- Mass Spring stiffness
- Damping coefficient

For every variation, both step and impulse responses were obtained and compared with the base configuration. This systematic approach enabled the individual contribution of each parameter to be identified without interference from simultaneous parameter changes.

7 Results and Discussion

Table 1 Description of Time Notation Used Throughout the Results and Discussion

Symbol	Description
t_1	Base case where ($m=1$), ($b=1$), and/or ($k=1$). This configuration serves as the reference for comparison.
t_2	Response obtained after increasing the mass while simultaneously increasing the spring stiffness or damping coefficient according to the selected case.
t_3	Response obtained when the spring stiffness or damping coefficient is increased more significantly than the mass.
t_{m0}	Response obtained while the mass remains constant and only the spring stiffness or damping coefficient is varied.
t_{k0}	Response obtained while the spring stiffness remains constant and only the mass is varied (Cases 1 and 3).
t_{b0}	Response obtained while the damping coefficient remains constant and only the mass is varied (Cases 2 and 4).

The mathematical models developed in the previous chapter were implemented in MATLAB to investigate the dynamic behavior of the six mechanical system configurations. The objective of this analysis was to understand how variations in mass, spring stiffness, and damping coefficient influence the transient characteristics of each system. Step and impulse responses were obtained for all configurations, while pole locations and analytical expressions were used to explain the observed behavior. The complete mass-spring-damper system was further analyzed using transient performance specifications and numerical solutions obtained through ODE45. The following sections discuss the engineering observations obtained from each case.

7.1 Case 1 — Mass–Spring System

7.1.1 System Description

The first configuration consists of a rigid mass connected to a single linear spring. Since no damping element is present, the system is incapable of dissipating energy. Consequently, any energy introduced into the system remains stored and continuously exchanges between the kinetic energy of the mass and the potential energy of the spring. The governing transfer function of this configuration is

$$G(s) = \frac{1}{ms^2 + k}$$

This represents a classical undamped second-order system.

7.1.2 Step Response Analysis

The step response demonstrates the behavior of the system when a constant force is suddenly applied and maintained.

The simulation results indicate that the mass oscillates continuously about its equilibrium position without any reduction in amplitude. This behavior occurs because the spring continuously converts kinetic energy into potential energy and vice versa, while no damping mechanism exists to remove energy from the system.

Two parameter studies were performed.

7.1.2.1 Constant Spring Stiffness

When the spring stiffness remained constant and the mass was increased, the oscillation period increased significantly. Physically, a larger mass possesses greater inertia, making it more difficult for the spring to accelerate the system. Consequently, the oscillations became slower, and the time required to complete one oscillation increased.

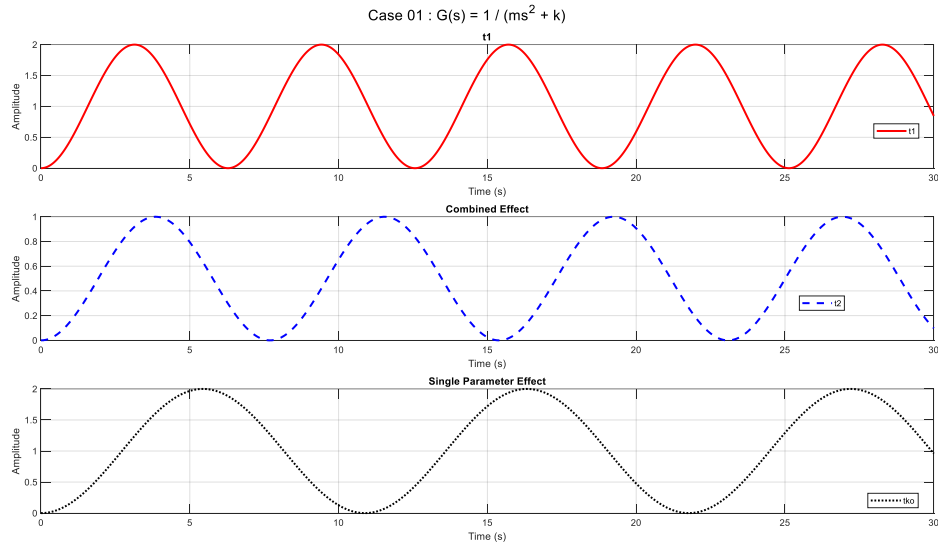


Figure 15 Step response plot, "constant spring stiffness" — Case 01

The observed trend

$$t_{k0} > t_2 > t_1$$

agrees with the theoretical expression

$$T = \frac{2\pi}{\omega_n}$$

where

$$\omega_n = \sqrt{\frac{k}{m}}$$

As the mass increases while the spring stiffness remains unchanged, the natural frequency decreases, resulting in a larger oscillation period.

7.1.2.2 Constant Mass

When the mass remained constant and the spring stiffness was increased, the oscillation period decreased.

A stiffer spring generates a larger restoring force for the same displacement, allowing the mass to return toward equilibrium more rapidly.

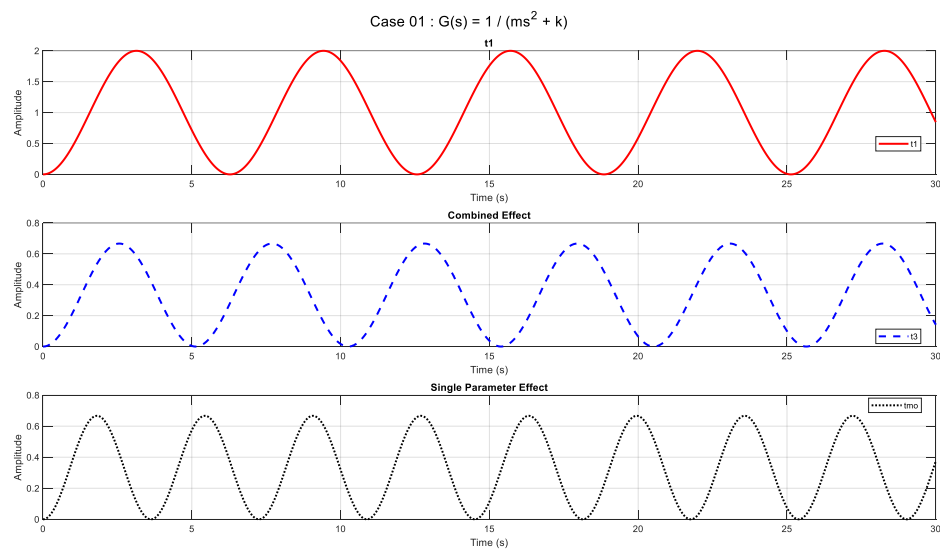


Figure 16 Step response plot, "constant mass" — Case 01

The simulation produced the trend

$$t_1 > t_3 > t_{m0}$$

which again agrees with the theoretical expression for the natural frequency.

Increasing the spring stiffness increases the natural frequency and therefore reduces the oscillation period.

7.1.3 Impulse Response Analysis

The impulse response exhibits similar oscillatory behavior to the step response because the system remains undamped.

Following the application of the impulse force, the mass acquires an initial velocity and continues oscillating indefinitely since no mechanism exists to dissipate the stored mechanical energy.

The same parameter trends observed in the step response were also obtained for the impulse response. Increasing the mass resulted in slower oscillations, whereas increasing the spring stiffness increased the oscillation frequency.

These observations further verify the theoretical relationship between the oscillation period and the natural frequency.

7.1.4 Pole Analysis

The poles of the mass-spring system lie entirely on the imaginary axis.

This indicates that the system is **marginally stable**, meaning the oscillations neither increase nor decrease with time.

Comparison of the different parameter variations revealed that increasing the spring stiffness moved the poles farther away from the origin along the imaginary axis. Consequently, the oscillation frequency increased and the oscillation period decreased.

```
t1 Poles
p1 =
    0.0000 + 1.0000i
    0.0000 - 1.0000i

t3 Poles
p3 =
    0.0000 + 1.2247i
    0.0000 - 1.2247i

tmo Poles
pmo =
    0.0000 + 1.7321i
    0.0000 - 1.7321i
```

Conversely, increasing the mass shifted the poles closer to the origin, reducing the natural frequency and increasing the oscillation period.

```
t1 Poles
p1 =
    0.0000 + 1.0000i
    0.0000 - 1.0000i

t2 Poles
p2 =
    0.0000 + 0.8165i
    0.0000 - 0.8165i

tko Poles
pko =
    0.0000 + 0.5774i
    0.0000 - 0.5774i
```

The pole locations therefore explain the trends observed in both the step and impulse responses.

7.1.5 Engineering Observation

The results demonstrate that the oscillatory behavior of an undamped system is governed entirely by the balance between inertia and stiffness. Increasing the mass slows the response due to greater inertia, whereas increasing the spring stiffness accelerates the response by generating a stronger restoring force. Since no damping mechanism exists, neither parameter is capable of eliminating the oscillations.

7.2 Case 02 – Mass–Damper System

7.2.1 System Description

The second configuration consists of a rigid mass connected to a single viscous damper without a spring element. Unlike the previous case, no restoring force exists to return the mass toward its equilibrium position. Instead, the damper continuously resists motion by producing a force proportional to the velocity of the mass.

The governing transfer function of this configuration is

$$G(s) = \frac{1}{ms^2 + bs}$$

This represents a first-order mechanical system in which energy is dissipated through viscous damping rather than stored and exchanged as in the mass-spring system.

7.2.2 Step Response Analysis

The step response represents the application of a constant external force to the mass. When the force is first applied, the mass begins to accelerate because the applied force is greater than the damping force. As the velocity increases, the damping force

$$F_d = bv$$

also increases.

Eventually, the damping force becomes equal to the applied force, resulting in

$$F_{applied} = F_d$$

At this instant, the net force acting on the mass becomes zero, causing the acceleration to vanish. Since no spring is present to oppose the displacement, the mass continues moving with a constant velocity. **Consequently, the displacement increases linearly with time, producing the straight-line response observed in the simulation.**

Two parameter studies were performed.

7.2.2.1 Constant Damping Coefficient

When the damping coefficient remained constant and the mass was increased, the response became slower. The larger inertia required more time for the velocity to reach its steady value.

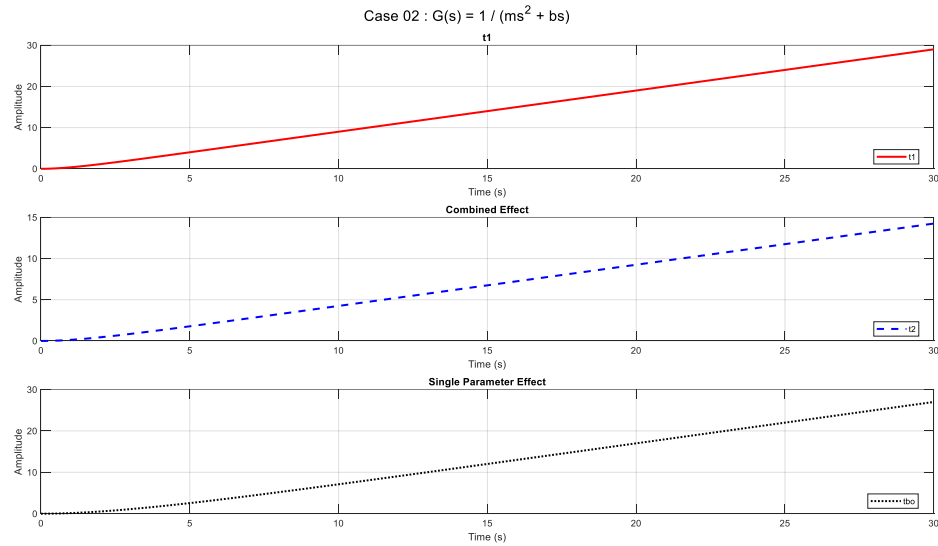


Figure 17 Step response plot, "constant damping coefficient" — Case 02

The observed trend

$$t_{b0} > t_2 > t_1$$

is consistent with the theoretical time constant

$$\tau = \frac{m}{b}$$

which shows that increasing the mass increases the response time of the system.

7.2.2.2 Constant Mass

When the mass remained constant and the damping coefficient was increased, the response became faster. A larger damping coefficient generated a greater resistive force for the same velocity, allowing the system to reach its constant velocity more quickly.

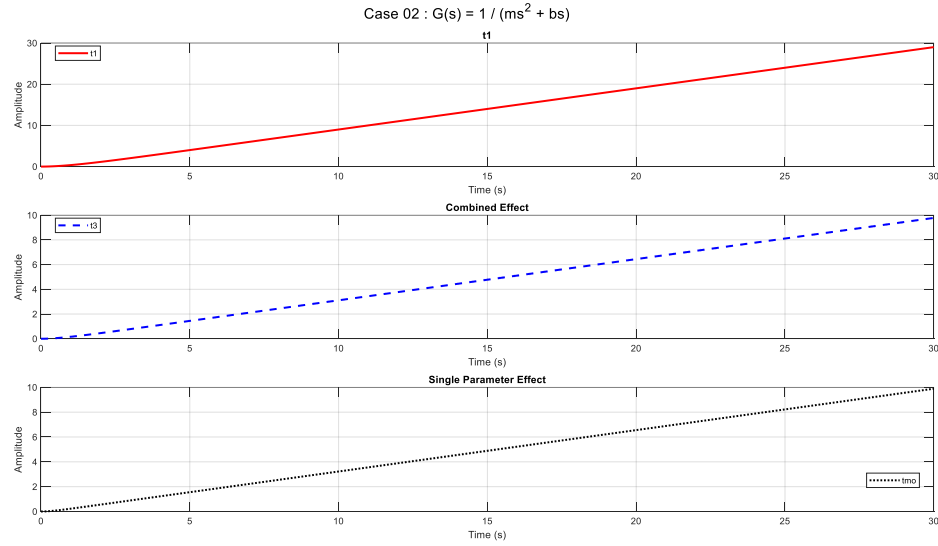


Figure 18 Step response plot, "constant mass" — Case 02

The simulation produced the trend

$$t_1 > t_3 > t_{m0}$$

which also agrees with

$$\tau = \frac{m}{b}$$

Since the damping coefficient appears in the denominator, increasing its value reduces the system time constant and produces a faster response.

7.2.3 Impulse Response Analysis

The impulse response provides a clearer understanding of the physical behavior of the mass-damper system.

Immediately after the impulse force is applied, the mass acquires an initial velocity. Once the impulse is removed, the applied force becomes zero, leaving the damping force as the only force acting on the system.

The damper continuously opposes the motion and gradually reduces the velocity until

$$v = 0$$

At this point, the damping force also becomes zero because

$$F_d = bv$$

Since no spring exists to restore the mass toward its original position, the mass remains at its final displacement. This behavior is observed in the impulse response, where the displacement initially increases before approaching a constant value without oscillations.

The steady displacement is determined by the initial impulse and the damping coefficient. Increasing the damping coefficient reduces the final displacement, whereas increasing the mass causes the system to require more time before coming to rest.

The parameter studies produced the same trends observed in the step response.

For constant damping,

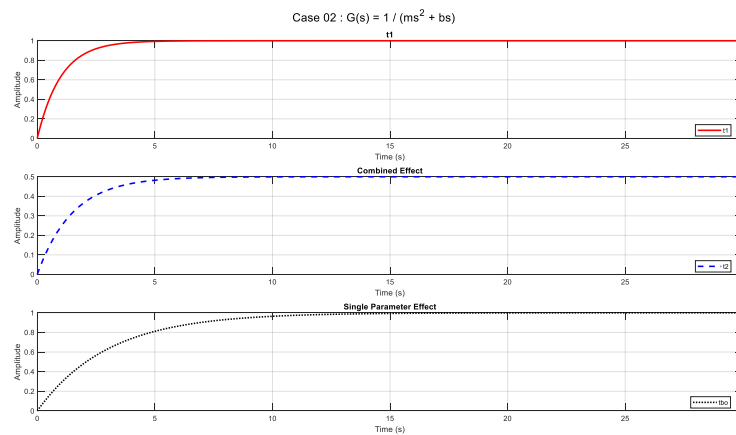


Figure 19 Impulse response plot, "constant damping" — Case 02

$$t_{b0} > t_2 > t_1$$

while for constant mass,

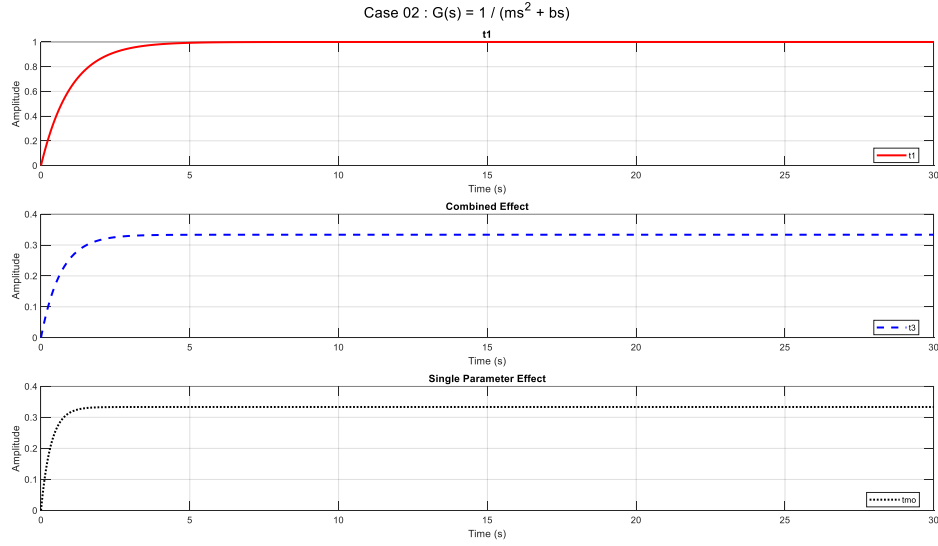


Figure 20 Impulse response plot, "constant mass" — Case 02

$$t_1 > t_3 > t_{m0}$$

These results confirm that increasing the mass slows the decay of motion, whereas increasing the damping coefficient accelerates the dissipation of kinetic energy.

7.2.4 Pole Analysis

The poles of the mass-damper system are located on the real axis, indicating that the response is non-oscillatory.

Unlike the mass-spring system, no complex conjugate poles are present because the absence of a spring prevents the storage and exchange of mechanical energy required for oscillations.

Comparison of the different parameter variations showed that increasing the damping coefficient moved the pole farther from the origin along the negative real axis, producing a faster response.

t1 Poles

p1 =

0
-1

t3 Poles

p3 =

0
-1.5000

tmo Poles

pmo =

0
-3

Conversely, increasing the mass shifted the pole closer to the origin, resulting in a slower decay of the response.

t1 Poles

p1 =

0
-1

t2 Poles

p2 =

0
-0.6667

tbo Poles

pbo =

0
-0.3333

These observations are fully consistent with the theoretical relationship

$$\tau = \frac{m}{b}$$

and explain the trends observed in both the step and impulse responses.

7.2.5 Engineering Observation

The results demonstrate that the behavior of a mass-damper system is governed entirely by inertia and viscous damping. Increasing the mass slows the system response because greater inertia resists changes in motion, whereas increasing the damping coefficient accelerates the dissipation of kinetic energy and reduces the time required for the velocity to decay to zero. Since no spring is present, the system cannot generate restoring forces or oscillatory motion. Consequently, all responses remain non-oscillatory, and after an impulse input the mass comes to rest at its final position rather than returning to its original equilibrium location.

7.3 Case 03 – Mass with Two Springs

7.3.1 System Description

The third configuration consists of a rigid mass connected to two identical linear springs attached on opposite sides of the mass. Since both springs oppose the displacement simultaneously, their stiffnesses combine to produce an equivalent spring stiffness

$$k_{eq} = 2k$$

The governing transfer function therefore becomes

$$G(s) = \frac{1}{ms^2 + 2k}$$

Similar to Case 01, no damping element is present. Consequently, the system remains undamped and continuously exchanges energy between the mass and the springs.

7.3.2 Step Response Analysis

The step response of the two-spring system exhibits the same oscillatory behavior observed in the single-spring configuration. However, because the equivalent stiffness is doubled, the restoring force acting on the mass becomes significantly larger for the same displacement.

This increased restoring force causes the mass to accelerate toward its equilibrium position more rapidly, producing oscillations with a higher natural frequency and a shorter oscillation period.

Two parameter studies were performed.

7.3.2.1 Constant Spring Stiffness

When the equivalent spring stiffness remained constant and the mass was increased, the oscillation period increased because the greater inertia reduced the natural frequency of the system.

The observed trend

$$t_{k0} > t_2 > t_1$$

remained identical to that obtained in Case 01 and is consistent with the theoretical relationship

$$T = \frac{2\pi}{\omega_n}$$

where

$$\omega_n = \sqrt{\frac{2k}{m}}$$

Increasing the mass decreases the natural frequency, resulting in slower oscillations.

7.3.2.2 Constant Mass

When the mass remained constant and the spring stiffness was increased, the oscillation period decreased due to the larger restoring force generated by the springs.

The observed trend

$$t_1 > t_3 > t_{m0}$$

again agrees with the natural frequency equation, demonstrating that increasing stiffness increases the oscillation frequency.

Although the parameter trends are identical to those observed in Case 01, the oscillation periods obtained in this configuration are noticeably shorter because the equivalent stiffness is twice that of the single-spring system.

7.3.3 Impulse Response Analysis

The impulse response exhibits continuous oscillations similar to those observed in the step response. After the impulse force is removed, the mechanical energy introduced into the system continues to alternate between the kinetic energy of the mass and the elastic potential energy stored in the springs.

Since no damping mechanism is available to dissipate this energy, the oscillations continue indefinitely.

Increasing the mass produced slower oscillations, whereas increasing the spring stiffness produced faster oscillations.

The same parameter trends observed in the step response were also obtained for the impulse response, confirming the validity of the theoretical model.

7.3.4 Pole Analysis

The poles of the two-spring system remain purely imaginary, indicating that the system is marginally stable and continues to oscillate indefinitely.

Compared with Case 01, the poles are located farther from the origin along the imaginary axis because of the increased equivalent stiffness.

The greater distance from the origin corresponds to a higher natural frequency, which explains the shorter oscillation periods observed in the simulations.

Table 2 Pole comparison table, Case 01 vs. Case 03 (varying stiffness, $t_1/t_3/t_{mo}$)

Parameter	Case 01 Poles	Case 03 Poles	Comparison
t_1	$\pm j1.0000$	$\pm j1.4142$	Case 03 poles are farther from the origin, indicating a higher natural frequency.
t_3	$\pm j1.2247$	$\pm j1.7321$	Both systems remain purely imaginary, but Case 03 has a larger pole magnitude.
t_{mo}	$\pm j1.7321$	$\pm j2.4495$	Case 03 exhibits the highest natural frequency due to the larger pole magnitude.

Similarly, increasing the mass shifts the poles closer to the origin, reducing the natural frequency and producing slower oscillations.

Table 3 Pole comparison table, Case 01 vs. Case 03 (varying mass, $t_1/t_2/t_{ko}$)

Parameter	Case 01 Poles	Case 03 Poles	Comparison
t_1	$\pm j1.0000$	$\pm j1.4142$	Case 03 poles are located farther from the origin, indicating a higher oscillation frequency.
t_2	$\pm j0.8165$	$\pm j1.1547$	Pole magnitude decreases in both cases, but Case 03 maintains higher-frequency dynamics.
t_{ko}	$\pm j0.5774$	$\pm j0.8165$	The poles move closer to the origin as stiffness decreases, while Case 03 still has larger pole magnitude than Case 01.

The pole locations therefore provide an accurate mathematical explanation for the dynamic behavior observed in both the step and impulse responses.

7.3.5 Comparison with Case 01

Although the trends observed in Case 03 are identical to those obtained in the single-spring system, the overall response is noticeably faster due to the increase in equivalent stiffness. For identical mass values, the two-spring configuration produces shorter oscillation periods than the single-spring configuration because

$$\omega_n = \sqrt{\frac{2k}{m}}$$

is greater than

$$\omega_n = \sqrt{\frac{k}{m}}$$

Consequently, oscillations occur more rapidly while preserving the same overall dynamic characteristics.

7.3.6 Engineering Observation

The addition of a second spring increases the overall stiffness of the mechanical system without changing its fundamental dynamic behavior. The system remains undamped and therefore continues to oscillate indefinitely. However, the higher equivalent stiffness increases the natural frequency, allowing the system to respond more quickly than the single-spring configuration. The parameter trends remain unchanged, indicating that the governing relationship between mass, stiffness, and natural frequency is preserved even when multiple springs are combined.

7.4 Case 04 – Mass with Two Dampers

7.4.1 System Description

The fourth configuration consists of a rigid mass connected to two identical viscous dampers attached on opposite sides of the mass. Since both dampers oppose the motion simultaneously, their damping coefficients combine to produce an equivalent damping coefficient

$$b_{eq} = 2b$$

The governing transfer function therefore becomes

$$G(s) = \frac{1}{ms^2 + 2bs}$$

Unlike the spring systems, this configuration contains no restoring element. Consequently, the system cannot oscillate and instead dissipates the applied mechanical energy through viscous damping.

7.4.2 Step Response Analysis

The step response of the two-damper system exhibits the same fundamental behavior as the single-damper configuration. Immediately after the step force is applied, the mass accelerates until the damping force increases sufficiently to balance the applied force.

When

$$F_{applied} = F_d$$

the net force acting on the mass becomes zero, causing the acceleration to vanish. Since no spring is present to return the mass toward its equilibrium position, the mass continues moving with a constant velocity, producing the linear displacement response observed in the simulations.

Two parameter studies were performed.

7.4.2.1 Constant Damping Coefficient

When the equivalent damping coefficient remained constant and the mass was increased, the response became slower because the increased inertia required more time for the velocity to reach its steady value.

The observed trend

$$t_{b0} > t_2 > t_1$$

is consistent with the theoretical time constant

$$\tau = \frac{m}{2b}$$

which indicates that increasing the mass increases the response time.

7.4.2.2 Constant Mass

When the mass remained constant and the damping coefficient was increased, the response became faster because the larger damping coefficient generated a greater resistive force for the same velocity.

The simulation produced the trend

$$t_1 > t_3 > t_{m0}$$

which also agrees with the theoretical time constant. Increasing the damping coefficient reduces the system time constant and allows the system to reach its steady velocity more rapidly.

7.4.3 Impulse Response Analysis

The impulse response demonstrates the dissipative nature of the two-damper system.

Immediately after the impulse force is applied, the mass gains an initial velocity. Once the external force is removed, the dampers become the only elements acting on the system. The viscous damping force continuously opposes the motion, gradually reducing the velocity until

$$v = 0$$

At this instant, the damping force also becomes zero because

$$F_d = 2bv$$

Since no spring exists to provide a restoring force, the mass remains at its final displacement after coming to rest. No oscillatory behavior is observed throughout the response because the system contains no energy-storing element.

The parameter variations produced the same trends observed in the step response. Increasing the mass delayed the decay of motion, whereas increasing the damping coefficient reduced the time required for the velocity to become zero.

7.4.4 Pole Analysis

The poles of the two-damper system remain on the negative real axis, confirming that the response is non-oscillatory.

Compared with the single-damper configuration (Case 02), the poles are located farther from the origin because the equivalent damping coefficient is doubled.

The increased distance from the origin corresponds to a faster dynamic response, explaining why the two-damper system reaches its steady condition more quickly than the single-damper system.

Table 4 Pole comparison table, Case 02 vs. Case 04 (varying damping)

Parameter	Case 02 Poles	Case 04 Poles	Comparison
t_1	0, -1	0, -2	Case 04 has a pole farther from the origin, resulting in a faster system response.
t_3	0, -1.5000	0, -3	The negative pole shifts farther left in Case 04, indicating increased response speed.
t_{mo}	0, -3	0, -6	Both systems become faster as mass decreases, with Case 04 exhibiting the fastest dynamics.

Increasing the mass shifts the poles toward the origin, producing a slower response, while increasing the damping coefficient moves the poles farther into the left-half plane, reducing the response time.

Table 5 Pole comparison table, Case 02 vs. Case 04 (varying mass)

Parameter	Case 02 Poles	Case 04 Poles	Comparison
t_1	0, -1	0, -2	Case 04 has a more negative pole, corresponding to a quicker transient response.
t_2	0, -0.6667	0, -1.3333	Both systems slow down as damping decreases, but Case 04 remains faster.
t_{bo}	0, -0.3333	0, -0.6667	The poles move closer to the origin as damping decreases, increasing the response time, while Case 04 still maintains a faster response than Case 02.

These observations agree closely with the theoretical time constant

$$\tau = \frac{m}{2b}$$

and validate the mathematical model developed for this configuration.

7.4.5 Comparison with Case 02

The overall dynamic behavior of the two-damper system is identical to that of the single-damper configuration because neither system contains a restoring spring.

However, doubling the damping coefficient reduces the system time constant, allowing the motion to decay more rapidly. Consequently, all response times obtained in Case 04 are smaller than those observed in Case 02, while the general response characteristics and parameter trends remain unchanged.

The pole locations further support this observation, as the poles of the two-damper system are positioned farther from the origin on the negative real axis, indicating a faster response than the corresponding single-damper system.

7.4.6 Engineering Observation

The addition of a second damper increases the rate at which mechanical energy is dissipated without altering the fundamental behavior of the system. Since no spring is present, the system cannot generate oscillations and therefore exhibits a purely non-oscillatory response. Increasing the equivalent damping coefficient produces faster decay of motion and shorter response times, whereas increasing the mass increases inertia and slows the system response. The trends observed in this configuration closely match those obtained for the single-damper system, with the primary difference being the reduction in response time caused by the increased equivalent damping coefficient.

7.5 Case 05 and Case 06 – Mass–Spring–Damper System

7.5.1 System Description

The fifth and sixth configurations represent the complete mass-spring-damper system. In Case 05, the spring and damper are connected on the same side of the mass, whereas in Case 06 they are connected on opposite sides. Despite the different physical arrangements, both systems possess the same governing equation of motion and therefore have the identical transfer function

$$G(s) = \frac{1}{ms^2 + bs + k}$$

Consequently, both configurations exhibit identical dynamic characteristics. Since the mathematical models are equivalent, only the results of Case 05 are discussed in detail. The conclusions obtained are equally applicable to Case 06.

7.5.2 Effect of Individual Parameter Variations

The complete second-order system was first analyzed by varying one physical parameter at a time while keeping the remaining parameters constant. This approach makes it possible to isolate the influence of mass, damping coefficient, and spring stiffness on the transient response.

The base case

$$m = 1, b = 1, k = 1$$

was selected as the reference configuration for all subsequent comparisons.

7.5.2.1 Base Case

“The base system exhibited the characteristic response of an underdamped second-order system.”

- For the **step input**, the displacement increased rapidly, exceeded the desired steady-state value, and then gradually settled to the final displacement after several oscillations.
- The **impulse response** similarly produced oscillations that decayed with time before eventually returning to the equilibrium position.

The **pole locations** consisted of a complex conjugate pair located in the left-half plane, confirming that the system is stable while simultaneously exhibiting oscillatory behavior.

This configuration serves as the reference against which all parameter variations are compared.

7.5.2.2 Effect of Increasing Mass

“The mass was increased while maintaining constant damping and spring stiffness.”

- The **step response** demonstrated a slower system response. Both the rise time and settling time increased because the larger inertia required greater time to accelerate and decelerate the system. The oscillations also became slower due to the reduction in natural frequency.

- The **impulse response** exhibited the same trend, requiring a longer time to return to equilibrium after the applied force was removed.

Pole analysis showed that increasing the mass moved the poles closer to the imaginary axis, reducing the natural frequency and slowing the overall system response.

These observations agree with the theoretical expression

$$\omega_n = \sqrt{\frac{k}{m}}$$

which indicates that increasing the mass decreases the natural frequency.

7.5.2.3 Effect of Increasing Spring Stiffness

“The spring stiffness was increased while maintaining constant mass and damping.”

- The **step response** became noticeably faster. The rise time and peak time decreased because the stronger restoring force accelerated the mass more effectively toward its equilibrium position.
- The **impulse response** returned to equilibrium more rapidly and exhibited higher-frequency oscillations.

Pole analysis showed that increasing the spring stiffness moved the complex poles farther from the origin along the imaginary direction, increasing the natural frequency of the system.

These observations are consistent with

$$\omega_n = \sqrt{\frac{k}{m}}$$

which predicts a higher natural frequency for larger spring stiffness.

7.5.2.4 Effect of Increasing Damping Coefficient

“The damping coefficient was increased while maintaining constant mass and spring stiffness.”

Unlike the previous parameter variations, increasing the damping coefficient primarily affected the oscillatory nature of the response rather than the natural frequency.

- The **step response** exhibited a significantly smaller overshoot and settled more smoothly toward its final value. Oscillations were rapidly suppressed because the damper dissipated mechanical energy more effectively.
- The **impulse response** also decayed much more rapidly, returning to equilibrium without prolonged oscillations.

Pole analysis showed that increasing the damping coefficient shifted the poles farther into the left-half plane, increasing the rate at which the transient response decayed.

These observations demonstrate that the damping coefficient primarily governs energy dissipation and therefore controls overshoot, oscillation amplitude, and settling characteristics.

7.5.2.5 Step Response

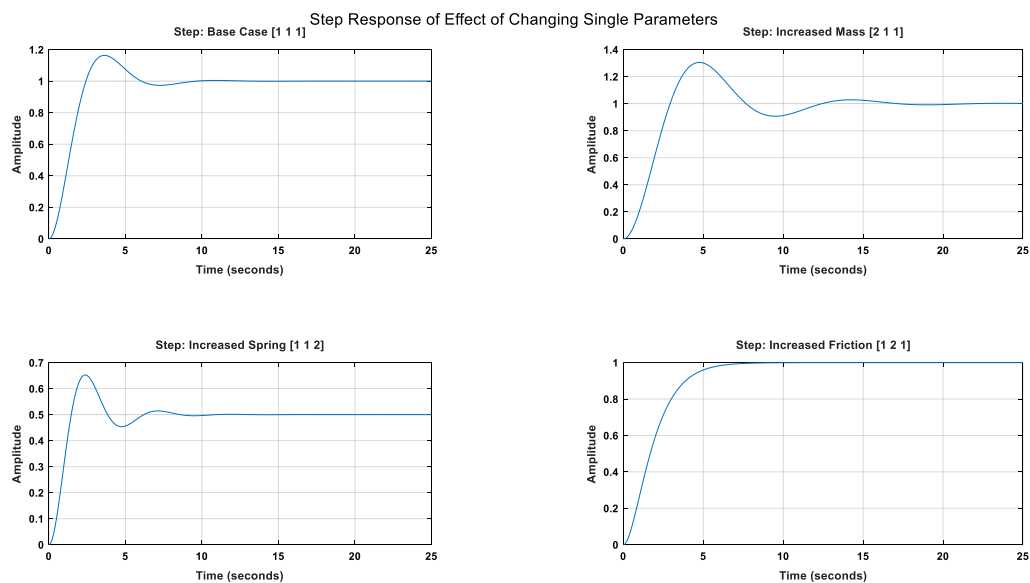


Figure 21 Step Response

7.5.2.6 Impulse Response

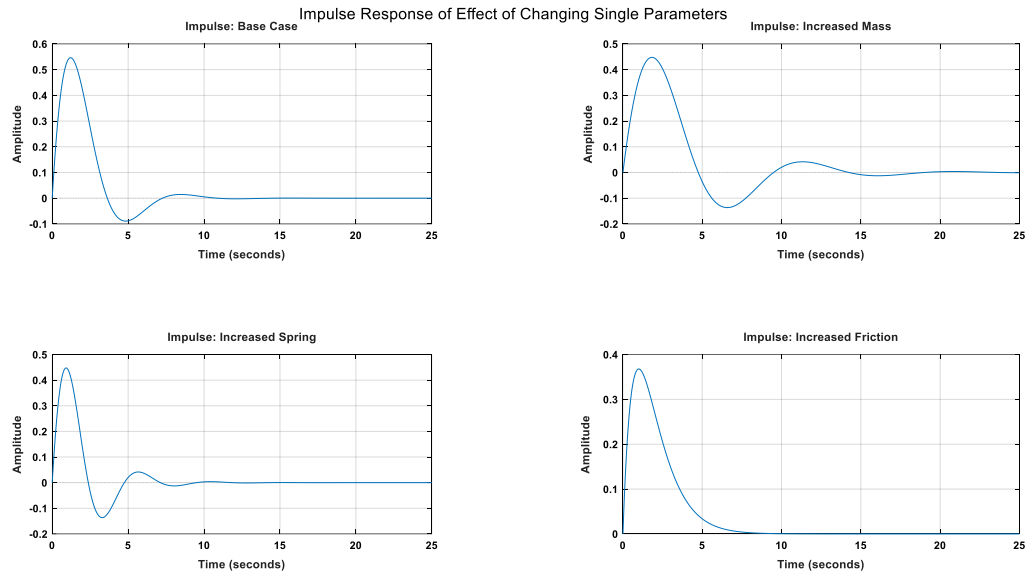


Figure 22 Impulse Response

7.5.3 Comparison of Underdamped and Overdamped Responses

To investigate the influence of the damping ratio, two additional parameter combinations were analyzed.

The first configuration satisfied

$$k > b > m$$

which produced an underdamped response.

The second configuration satisfied

$$b > k > m$$

which produced an overdamped response.

- The **underdamped system** responded rapidly but exhibited oscillations and overshoot before reaching steady state. The corresponding poles formed a complex conjugate pair in the left-half plane, confirming oscillatory behavior.
- In contrast, the **overdamped system** exhibited a slower response with no oscillations or overshoot. The poles became purely real and negative, indicating

that the response was dominated by exponential decay rather than oscillatory motion.

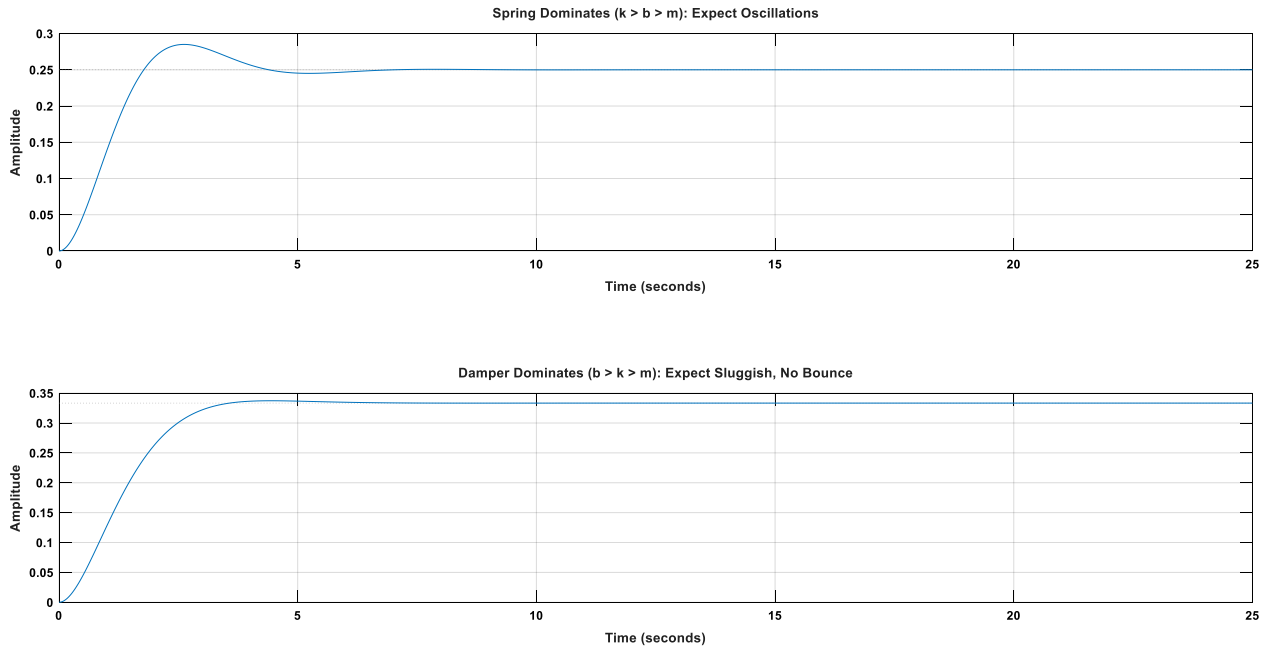


Figure 23 Overdamped vs. underdamped decay plot

Comparison of the transient performance specifications showed that the underdamped system achieved a faster initial response, whereas the overdamped system required a longer time to reach steady state but eliminated oscillations almost entirely.

These results illustrate the fundamental engineering trade-off between response speed and oscillatory behavior in second-order control systems.

7.5.4 Pole Analysis

The pole locations accurately explained every trend observed during the simulations.

Increasing the mass shifted the poles toward the imaginary axis, producing slower responses.

Increasing the spring stiffness increased the natural frequency by moving the poles farther from the origin along the imaginary direction.

Increasing the damping coefficient shifted the poles farther into the left-half plane, reducing oscillations and improving stability.

Table 6 Pole comparison table, condition-based

Condition	System Poles
Base Case	$-0.5000 \pm 0.8660i$
Increased Mass	$-0.2500 \pm 0.6614i$
Increased Spring Constant	$-0.5000 \pm 1.3229i$
Increased Friction	-1, -1
Underdamped Case	$-0.7500 \pm 1.1990i$
Overdamped Case	$-1.0000 \pm 0.7071i$

Furthermore, comparison of the underdamped and overdamped systems clearly demonstrated the relationship between pole locations and transient response characteristics. Complex conjugate poles produced oscillatory behavior, whereas real negative poles generated smooth, non-oscillatory responses.

7.5.5 Performance Specification Analysis

Transient performance specifications obtained using the MATLAB `stepinfo` function further quantified the observed system behavior.

The measured rise time, settling time, peak time, and percentage overshoot confirmed the trends identified through visual inspection of the response curves.

- **Increasing the mass** increased both the rise time and settling time because of the greater inertia.
- **Increasing the spring stiffness** reduced the rise time and peak time by increasing the restoring force acting on the mass.
- **Increasing the damping coefficient** significantly reduced the overshoot and suppressed oscillations, resulting in a smoother transient response.

Table 7 Transient performance table—Base, Increased Mass, Increased Spring, Increased Friction

Case	Rise Time (s)	Peak Time (s)	Settling Time (s)	Overshoot (%)
Base	1.64	3.59	8.08	16.29
Increased Mass	1.98	4.79	15.48	30.49
Increased Spring	0.99	2.39	7.74	30.49
Increased Friction	3.36	9.79	5.83	0.00

Comparison of the underdamped and overdamped systems further demonstrated that larger damping ratios reduce oscillatory behavior at the expense of slower response speed.

- The underdamped system reaches the desired value faster.
- The overdamped system settles sooner despite having a slower rise time.

- The underdamped system exhibits significantly higher overshoot, whereas the overdamped system has a nearly overshoot-free response.

Table 8 Transient performance table — Underdamped vs. Overdamped comparison

Case	Rise Time (s)	Peak Time (s)	Settling Time (s)	Overshoot (%)
Underdamped	1.20	2.64	4.09	14.01
Overdamped	2.07	4.42	3.18	1.18

7.5.6 ODE45 Verification

To verify the transfer-function-based analysis, the governing differential equation of the base mass-spring-damper system was solved numerically using the MATLAB `ode45` solver.

The displacement response obtained from the numerical solution closely matched the response generated using the transfer function. Both approaches produced identical transient characteristics, including the oscillatory behavior, steady-state value, and settling characteristics.

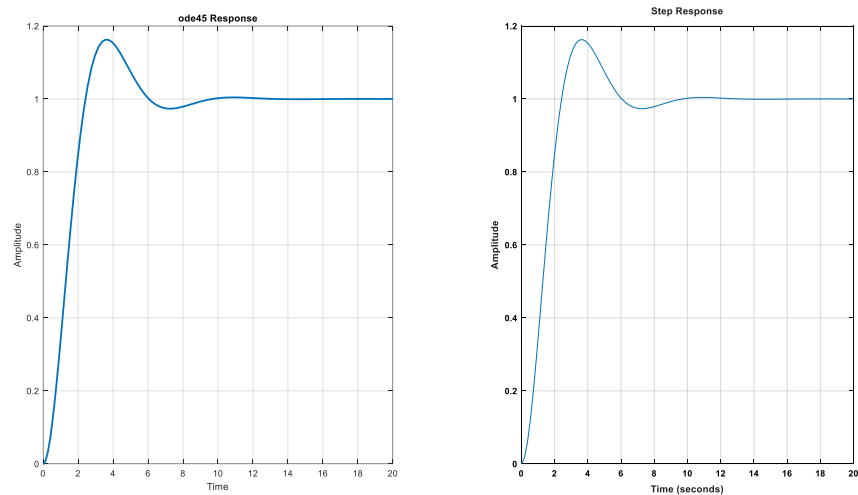


Figure 24 ODE45 numerical vs. transfer-function verification plot

This agreement verifies that the mathematical model, transfer function, and numerical solution accurately describe the physical dynamics of the mass-spring-damper system.

7.5.7 Engineering Observation

The complete mass-spring-damper system combines the effects of inertia, restoring force, and energy dissipation into a single second-order dynamic model. Among the three physical parameters, the mass primarily governs the speed of the response through its influence on inertia, the spring determines the natural frequency and restoring capability of the system, while the damper controls the decay of oscillations and overall stability. The study further demonstrates that system poles provide a direct mathematical interpretation of the observed dynamic behavior, allowing transient characteristics to be predicted from the transfer function alone. The close agreement between the transfer-function simulations and the ODE45 numerical solution validates both the mathematical model and the simulation methodology employed throughout the project.

8 Aerospace Relation

Although the mass-spring-damper system is a simplified mechanical model, it forms the foundation of many aerospace engineering applications. Numerous aircraft and spacecraft components can be approximated as second-order dynamic systems, making the analysis performed in this project directly relevant to aerospace control engineering.

8.1 Aircraft Application

In aircraft, landing gear suspension systems behave as mass-spring-damper systems, where the aircraft mass is supported by shock absorbers and springs that absorb the impact energy during landing while minimizing oscillations. Similarly, aircraft pitch, roll, and yaw dynamics can often be approximated by second-order models when designing automatic flight control systems.

8.2 Space Vehicles Application

In spacecraft and satellites, flexible solar panels, antennas, and structural components exhibit vibration characteristics that closely resemble mass-spring-damper systems. Understanding these dynamics is essential for preventing structural oscillations and maintaining pointing accuracy during missions.

8.3 UAVs Application

The concepts investigated in this project are also applicable to unmanned aerial vehicles (UAVs), where controllers continuously adjust the control surfaces or rotor speeds to stabilize the vehicle against external disturbances such as wind gusts. The relationship between pole locations, damping ratio, natural frequency, and transient response plays a fundamental role in designing stable and responsive flight control systems.

Furthermore, the **MATLAB and Simulink** techniques employed throughout this project are widely used in the aerospace industry for dynamic system modeling, controller design, simulation, verification, and performance evaluation prior to hardware implementation.

This project therefore provides the theoretical and computational foundation required for the study of advanced aerospace control systems, including aircraft autopilots, missile

guidance, launch vehicle attitude control, satellite attitude determination and control systems (ADCS), and autonomous UAV navigation.

9 Conclusions

This project investigated the dynamic behavior of six mechanical system configurations using transfer function modeling and MATLAB simulations. Each configuration was analyzed through step and impulse responses to understand the influence of mass, spring stiffness, and damping coefficient on system dynamics.

The mass-spring system demonstrated continuous oscillatory behavior due to the absence of damping, while the mass-damper system produced non-oscillatory responses because mechanical energy was dissipated through viscous damping. The equivalent two-spring and two-damper configurations exhibited the same dynamic characteristics as their single-element counterparts, although the increased equivalent stiffness and damping coefficients resulted in faster system responses.

The complete mass-spring-damper system provided a comprehensive understanding of second-order dynamics. Parameter variation studies showed that increasing the mass slowed the response, increasing the spring stiffness increased the natural frequency, and increasing the damping coefficient reduced oscillations and overshoot while improving system stability. Pole analysis successfully explained the observed transient behavior, demonstrating the direct relationship between pole locations and system performance.

Transient performance specifications obtained using MATLAB further quantified the influence of parameter variations, while the numerical solution obtained using the ODE45 solver closely matched the transfer-function-based simulations, validating the mathematical model developed throughout the project.

Overall, this project established a strong theoretical and practical understanding of classical control system fundamentals, MATLAB-based system analysis, and second-order dynamic behavior. The knowledge gained serves as a foundation for the study of advanced controller design techniques and aerospace control applications.

10 Future Work

The present project focused on open-loop dynamic system analysis using classical control system techniques. Several extensions can be implemented in future work to further improve the system performance and increase the complexity of the developed models.

Future work may include the design and tuning of **proportional (P), proportional-integral (PI), proportional-derivative (PD), and proportional-integral-derivative (PID) controllers** to investigate their effects on transient response, steady-state error, and overall system stability. Closed-loop feedback control can then be implemented in Simulink to compare the performance of controlled and uncontrolled systems.

The mechanical models developed in MATLAB may also be recreated using **Simscape Multibody** to produce realistic physical simulations and animations. Such models would provide a closer representation of actual engineering systems and enable visualization of component motion during dynamic response.

Further studies may include frequency-domain analysis using Bode plots, Nyquist plots, and Root Locus techniques, as well as state-space modeling, controllability and observability analysis, and modern control system design methods such as **Linear Quadratic Regulator (LQR), Model Predictive Control (MPC), and Robust Control**.

Finally, the methodology developed in this project can be extended to real aerospace applications, including aircraft longitudinal and lateral dynamics, UAV stabilization, rocket attitude control, satellite attitude control systems, and active vibration suppression of aerospace structures. Integration with sensors, actuators, and embedded hardware would enable the transition from simulation-based studies to real-time control system implementation.

“This project serves as the first stage of a broader control systems learning roadmap, where progressively more advanced projects involving Simulink, PID controller design, state-space modeling, modern control techniques, and aerospace flight control systems will be developed in subsequent studies.”

11 Appendix

11.1 MATLAB Code

```
clc , clear all , close all
```

```
%% Mass Spring Damper System with 6 cases
```

```
%Parameters of the system  
% m = 2 ; % Mass of block  
% k = 1 ; % Spring coefficient  
% b = 1 ; % Damping coefficient
```

11.1.1 Case 01

```
                                Case 01  
%                                 $G(s) = 1 / (ms^2 + k)$   
  
% num1 = 1;  
% den1 = [m 0 k];  
% G1 = tf(num1,den1);  
  
num1 = 1;  
  
% ----- t1 -----  
% m = 1 , k = 1  
den1_1 = [1 0 1];  
  
% ----- Combined Effect -----  
  
% t2 : Increase Mass and Spring Constant  
% den1_2 = [3 0 2];  
  
% t3 : Increase Spring Constant and Mass  
den1_2 = [2 0 3];  
  
% ----- Single Parameter Effect -----  
  
% tmo : Only Spring Constant Changes (m constant)  
den1_3 = [1 0 3];  
  
% tko : Only Mass Changes (k constant)  
% den1_3 = [3 0 1];
```

```

% Transfer Functions

G1_1 = tf(num1,den1_1);
disp('t1 Poles')
p1 = pole(G1_1)

G1_2 = tf(num1,den1_2);
disp('t3 Poles')
p3 = pole(G1_2)

G1_3 = tf(num1,den1_3);
disp('tmo Poles')
pmo = pole(G1_3)

% Time

t = 0:0.01:30;

% Responses

[y1s,t1s] = step(G1_1,t);
[y2s,t2s] = step(G1_2,t);
[y3s,t3s] = step(G1_3,t);

% [y1i,t1i] = impulse(G1_1,t);
% [y2i,t2i] = impulse(G1_2,t);
% [y3i,t3i] = impulse(G1_3,t);

% Plot

figure(1)

subplot(3,1,1)

plot(t1s,y1s,'r','LineWidth',2)
% plot(t1i,y1i,'r','LineWidth',2)

title('t1')
xlabel('Time (s)')
ylabel('Amplitude')
legend('t1','Location','best')
grid on

subplot(3,1,2)

```

```

plot(t2s,y2s,'b--','LineWidth',2)
% plot(t2i,y2i,'b--','LineWidth',2)

title('Combined Effect')
xlabel('Time (s)')
ylabel('Amplitude')

% legend('t2','Location','best')
legend('t3','Location','best')

grid on

subplot(3,1,3)

plot(t3s,y3s,'k:','LineWidth',2)
% plot(t3i,y3i,'k:','LineWidth',2)

title('Single Parameter Effect')
xlabel('Time (s)')
ylabel('Amplitude')

% legend('tko','Location','best')
legend('tmo','Location','best')

grid on

sgtitle('Case 01 :  $G(s) = 1 / (ms^2 + k)$ ')

```

11.1.2 Case 02

```

%%                                Case 02
%                                 $G(s) = 1 / (ms^2 + bs)$ 

% num2 = 1;
% den2 = [m b 0];
% G2 = tf(num2,den2);

num2 = 1;

% ----- t1 -----
% m = 1 , b = 1
den2_1 = [1 1 0];

% ----- Combined Effect -----

```

```

% t2 : Increase Mass and Damping
% den2_2 = [3 2 0];

% t3 : Increase Damping and Mass
den2_2 = [2 3 0];

% ----- Single Parameter Effect -----

% tmo : Only Damping Changes (m constant)
den2_3 = [1 3 0];

% tbo : Only Mass Changes (b constant)
% den2_3 = [3 1 0];

% Transfer Functions

G2_1 = tf(num2,den2_1);
disp('t1 Poles')
p1 = pole(G2_1)

G2_2 = tf(num2,den2_2);
disp('t3 Poles')
p3 = pole(G2_2)

G2_3 = tf(num2,den2_3);
disp('tmo Poles')
pmo = pole(G2_3)

% Time

t = 0:0.01:30;

% Responses
%
[y1s,t1s] = step(G2_1,t);
[y2s,t2s] = step(G2_2,t);
[y3s,t3s] = step(G2_3,t);

% [y1i,t1i] = impulse(G2_1,t);
% [y2i,t2i] = impulse(G2_2,t);
% [y3i,t3i] = impulse(G2_3,t);

% Plot

figure(2)

```

```

subplot(3,1,1)

plot(t1s,y1s,'r','LineWidth',2)
% plot(t1i,y1i,'r','LineWidth',2)

title('t1')
xlabel('Time (s)')
ylabel('Amplitude')
legend('t1','Location','best')
grid on

subplot(3,1,2)

plot(t2s,y2s,'b--','LineWidth',2)
% plot(t2i,y2i,'b--','LineWidth',2)

title('Combined Effect')
xlabel('Time (s)')
ylabel('Amplitude')

% legend('t2','Location','best')
legend('t3','Location','best')

grid on

subplot(3,1,3)

plot(t3s,y3s,'k:','LineWidth',2)
% plot(t3i,y3i,'k:','LineWidth',2)

title('Single Parameter Effect')
xlabel('Time (s)')
ylabel('Amplitude')

legend('tmo','Location','best')
% legend('tbo','Location','best')

grid on

sgtitle('Case 02 :  $G(s) = 1 / (ms^2 + bs)$ ')

```


11.1.3 Case 03

```
%%                                Case 03
%                                 $G(s) = 1 / (ms^2 + 2k)$ 

% num3 = 1;
% den3 = [m 0 2*k];
% G3 = tf(num3,den3);

num3 = 1;

% ----- t1 -----
% m = 1 , 2k = 2
den3_1 = [1 0 2];

% ----- Combined Effect -----

% t2 : Increase Mass and Spring Constant
% den3_2 = [3 0 4];

% t3 : Increase Spring Constant and Mass
den3_2 = [2 0 6];

% ----- Single Parameter Effect -----

% tmo : Only Spring Constant Changes (m constant)
den3_3 = [1 0 6];

% tko : Only Mass Changes (k constant)
% den3_3 = [3 0 2];

% Transfer Functions

G3_1 = tf(num3,den3_1);
disp('t1 Poles')
p1 = pole(G3_1)

G3_2 = tf(num3,den3_2);
disp('t3 Poles')
p3 = pole(G3_2)

G3_3 = tf(num3,den3_3);
disp('tmo Poles')
pmo = pole(G3_3)
```

11.1.4 Case 04

```
%%                                Case 04
%                                 $G(s) = 1 / (ms^2 + 2bs)$ 

% num4 = 1;
% den4 = [m 2*b 0];
% G4 = tf(num4,den4);

num4 = 1;

% ----- t1 -----
% m = 1 , 2b = 2
den4_1 = [1 2 0];

% ----- Combined Effect -----

% t2 : Increase Mass and Damping
% den4_2 = [3 4 0];

% t3 : Increase Damping and Mass
den4_2 = [2 6 0];

% ----- Single Parameter Effect -----

% tmo : Only Damping Changes (m constant)
den4_3 = [1 6 0];

% tbo : Only Mass Changes (b constant)
% den4_3 = [3 2 0];

% Transfer Functions

G4_1 = tf(num4,den4_1);
disp('t1 Poles')
p1 = pole(G4_1)

G4_2 = tf(num4,den4_2);
disp('t3 Poles')
p3 = pole(G4_2)

G4_3 = tf(num4,den4_3);
disp('tmo Poles')
pmo = pole(G4_3)
```

11.1.5 Case 05

```
                                %% Case 05

num5 = 1;
den5 = [m b k];
G5 = tf(num5,den5);

num5 = 1;
t = 0:0.01:25;

num5 = 1;
t = 0:0.01:25;

%% --- FIGURE 1: Changing One Parameter at a Time ---
figure(1)

% Base Case Step [1 1 1]
G5_1 = tf(num5, [1 1 1]);
subplot(2,2,1); step(G5_1, t); title('Step: Base Case [1 1 1]');
grid on;
disp('Base Case Poles')
p1 = pole(G5_1)

% Increased Mass Step [2 1 1]
G5_2 = tf(num5, [2 1 1]);
subplot(2,2,2); step(G5_2, t); title('Step: Increased Mass [2 1 1]'); grid on;
disp('Increased Mass Poles')
p2 = pole(G5_2)

% Increased Spring Step [1 1 2]
G5_3 = tf(num5, [1 1 2]);
subplot(2,2,3); step(G5_3, t); title('Step: Increased Spring [1 1 2]'); grid on;
disp('Increased Spring Poles')
p3 = pole(G5_3)

% Increased Friction Step [1 2 1]
G5_4 = tf(num5, [1 2 1]);
subplot(2,2,4); step(G5_4, t); title('Step: Increased Friction [1 2 1]'); grid on;
disp('Increased Friction Poles')
p4 = pole(G5_4)

sgtitle('Step Response of Effect of Changing Single Parameters')
```

```

figure(2)
% Base Case Impulse
subplot(2,2,1); impulse(G5_1, t); title('Impulse: Base Case');
grid on;

% Increased Mass Impulse
subplot(2,2,2); impulse(G5_2, t); title('Impulse: Increased
Mass'); grid on;

% Increased Spring Impulse
subplot(2,2,3); impulse(G5_3, t); title('Impulse: Increased
Spring'); grid on;

% Increased Friction Impulse
subplot(2,2,4); impulse(G5_4, t); title('Impulse: Increased
Friction'); grid on;

sgtitle('Impulse Response of Effect of Changing Single
Parameters')

info_B = stepinfo(G5_1);
fprintf('\n--- Base Case [1 1 1] Properties ---\n');
fprintf('Rise Time: %.2f seconds\n', info_B.RiseTime);
fprintf('Peak Time: %.2f seconds\n', info_B.PeakTime);
fprintf('Settling Time: %.2f seconds\n', info_B.SettlingTime);
fprintf('Overshoot: %.2f%%\n', info_B.Overshoot);

info_M = stepinfo(G5_2);
fprintf('\n--- Increased Mass Case [2 1 1] Properties ---\n');
fprintf('Rise Time: %.2f seconds\n', info_M.RiseTime);
fprintf('Peak Time: %.2f seconds\n', info_M.PeakTime);
fprintf('Settling Time: %.2f seconds\n', info_M.SettlingTime);
fprintf('Overshoot: %.2f%%\n', info_M.Overshoot);

info_S = stepinfo(G5_3);
fprintf('\n--- Increased Spring Case [1 1 2] Properties ---\n');
fprintf('Rise Time: %.2f seconds\n', info_S.RiseTime);
fprintf('Peak Time: %.2f seconds\n', info_S.PeakTime);
fprintf('Settling Time: %.2f seconds\n', info_S.SettlingTime);
fprintf('Overshoot: %.2f%%\n', info_S.Overshoot);

info_F = stepinfo(G5_4);
fprintf('\n--- Increased Friction Case [1 2 1] Properties ---
\n');
fprintf('Rise Time: %.2f seconds\n', info_F.RiseTime);

```

```

fprintf('Peak Time: %.2f seconds\n', info_F.PeakTime);
fprintf('Settling Time: %.2f seconds\n', info_F.SettlingTime);
fprintf('Overshoot: %.2f%%\n', info_F.Overshoot);

%% --- FIGURE 2: When Damping Dominates vs. Spring Dominates ---
figure('Name', 'System Inequality Comparisons');

% t5:  $k > b > m$  [2 3 4] -> Highly oscillatory (Underdamped)
G5_5 = tf(num5, [2 3 4]);
disp('Underdamped Poles')
p5 = pole(G5_5)
subplot(2,1,1); step(G5_5, t); grid on;
title('Spring Dominates ( $k > b > m$ ): Expect Oscillations');

% t6:  $b > k > m$  [2 4 3] -> Sluggish, slow rise (Overdamped)
G5_6 = tf(num5, [2 4 3]);
disp('Overdamped Poles')
p6 = pole(G5_6)
subplot(2,1,2); step(G5_6, t); grid on;
title('Damper Dominates ( $b > k > m$ ): Expect Sluggish, No Bounce');

info_U = stepinfo(G5_5);
fprintf('\n--- Underdamped Case [2 3 4] Properties ---\n');
fprintf('Rise Time: %.2f seconds\n', info_U.RiseTime);
fprintf('PeakTime: %.2f seconds\n', info_U.PeakTime);
fprintf('Settling Time: %.2f seconds\n', info_U.SettlingTime);
fprintf('Overshoot: %.2f%%\n', info_U.Overshoot);

info_O = stepinfo(G5_6);
fprintf('\n--- Overdamped Case [2 4 3] Properties ---\n');
fprintf('Rise Time: %.2f seconds\n', info_O.RiseTime);
fprintf('Peak Time: %.2f seconds\n', info_O.PeakTime);
fprintf('Settling Time: %.2f seconds\n', info_O.SettlingTime);
fprintf('Overshoot: %.2f%%\n', info_O.Overshoot);

```

11.1.6 Ode45 Solver

```
m = 1 ;
k = 1 ;
b = 1 ;
F = 1 ;

f = @(t,X) [X(2)
            (F-b*X(2)-k*X(1))/m] ;
tSpan = [0 20];
X0 = [0 0];
[t,X] = ode45(f,tSpan,X0);

subplot(1,2,1)
plot(t,X(:,1),"LineWidth",2)
grid on
xlabel("Time") ; ylabel("Amplitude")
title("ode45 Response")

subplot(1,2,2)
G = tf(1,[1 1 1]);
step(G,tSpan)
```